

JEE MAIN REVISION MASTER: SURFACE CHEMISTRY 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Surface Chemistry - Core Application	19 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Surface Chemistry - Core Application (Very High Yield — 19 PYQs)

Core Principle: Key problem pattern in Kcet Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the incorrect statement..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "For the coagulations of positively charged hydrated ferric - oxide sol, the flocculating power of the ions is in the order:..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Gold sol is not a..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Identify the incorrect statement

- (A) Values of colligative properties of colloidal solution are of small order compared to values of true solution.
- (B) Tyndall effect is observed only when diameter of the dispersed particles is not much smaller than wavelength of incident light.
- (C) Colour of colloidal solution depends on the wavelength of light scattered by the dispersed particles.
- (D) Brownian movement is due to balanced bombardment of molecules of dispersion medium on colloidal particles.

Correct Answer: None

Solution Analysis: Among the given statements incorrect statement is given in option (d). Its correct form is "Brownian movement is due to unbalanced bombardment of molecules of dispersion medium on colloidal particles".